

Commerce XI Maths App

Exercise 1.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 18

Question 1 - Q1(i)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63
Calculate D_6 .

Correct answer: 69.2

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 69.2.

Question 2 - Q1(ii)

For data : 79, 82, 36, 38, 51, 72, 68, 70, 64, 63
Calculate P_{85} .

Correct answer: 80.05

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 80.05.

Question 3 - Q2(i)

The daily wages (in ₹) of 15 labours are
230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250
Calculate D_8 .

Correct answer: 379

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 379.

Question 4 - Q2(ii)

The daily wages (in ₹) of 15 labours are
230, 400, 350, 200, 250, 380, 210, 225, 375, 180, 375, 450, 300, 350, 250
Calculate P_{90} .

Correct answer: 420

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 420.

Question 5 - Q3(i)

Calculate 2nd decile for the following :

x	80	100	120	145	200	280	310	380	400	410
f	15	18	25	27	40	25	19	16	8	7

Correct answer: 120

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 120.

Question 6 - Q3(ii)

Calculate 65th percentile for the following :

x	80	100	120	145	200	280	310	380	400	410
f	15	18	25	27	40	25	19	16	8	7

Correct answer: 280

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 280.

Question 7 - Q4(i)

From the following data calculate the rent of 15th house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 11000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 11000.

Question 8 - Q4(ii)

From the following data calculate the rent of 65th house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 14000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14000.

Question 9 - Q4(iii)

From the following data calculate the rent of 91st house.

House Rent (in ₹)	11000	12000	13000	14000	15000	16000	17000	18000
No. of houses	25	17	13	14	15	8	6	2

Correct answer: 16000

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 16000.

Question 10 - Q5(i)

The following frequency distribution shows the weight of students in a class.

Weight (in kg)	40	45	50	55	60	65
Number of Students	15	40	29	21	10	5

Find the percentage of students whose weight is more than 50 kg.

Correct answer: 30

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 30.

Question 11 - Q5(ii)

The following frequency distribution shows the weight of students in a class.

Weight (in kg)	40	45	50	55	60	65
Number of Students	15	40	29	21	10	5

If the weight column provided is of mid values then,
find the percentage of students whose weight is more than 50 kg.

Correct answer: 42

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 42.

Question 12 - Q6(i)

Calculate D_4 from the following data:

Mid Value	2.5	7.5	12.5	17.5	22.5	Total
Frequency	7	18	25	30	20	100

Correct answer: 13

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 13.

Question 13 - Q6(ii)

Calculate P_{48} from the following data:

Mid Value	2.5	7.5	12.5	17.5	22.5	Total
Frequency	7	18	25	30	20	100

Correct answer: 14.6

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 14.6.

Question 14 - Q7(i)

Calculate D_9 of the following distribution.

Length (in Inches)	0-20	20-40	40-60	60-80	80-100	100-120
No. of units	1	14	35	85	90	15

Correct answer: 98

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 98.

Question 15 - Q7(ii)

Calculate P_{20} of the following distribution.

Length (in Inches)	0-20	20-40	40-60	60-80	80-100	100-120
No. of units	1	14	35	85	90	15

Correct answer: 58.86

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 58.86.

Question 16 - Q8

Weekly wages for group of 100 persons are given below :

Weekly wages (in Rs.)	0-500	500-1000	1000-1500	1500-2000	2000-2500
No. of persons	7	a	25	30	b

D_3 for this group is Rs. 1100. Calculate the missing frequencies a, b.

Options

A. (18, 20)

B. (20, 18)

C. (17, 21)

D. (22, 16)

Correct answer: (18, 20)

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is (18, 20).

Question 17 - Q9

The weekly profit (in rupees) of 100 shops are distributed as follows :

Profit per shop	0-1000	1000-2000	2000-3000	3000-4000	4000-5000	5000-6000	6000-7000
No. of Shops	10	16	26	20	20	5	3

Find the limits of the profit of middle 60% of the shops.

Options

A. $(P_{20}, P_{80}) = (1625, 4400)$

B. $(P_{20}, P_{80}) = (1500, 4500)$

C. $(P_{30}, P_{70}) = (1625, 4400)$

D. $(P_{20}, P_{80}) = (1600, 4600)$

Correct answer: $(P_{20}, P_{80}) = (1625, 4400)$

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is $(P_{20}, P_{80}) = (1625, 4400)$.

Question 18 - Q10

In a particular factory, workers produce various types of output units.
The following distribution was obtained

Output units Produced	70-74	75-79	80-84	85-89	90-94	95-99	100-104
No. of workers	40	45	50	60	70	80	100

Find the percentage of workers who have produced less than 82 output units.

Correct answer: 24.72

Explanation and steps

Use $i(n+1)/10$ or $i(n+1)/100$ for raw data, and the corresponding grouped-data interpolation formula when classes are given. The result is 24.72.